

Micro Hack 2 — SDC Workloads

Participant Workbook · Fabric Extensibility Toolkit (GreenGrid)

FY27 EMEA EPS — Fabric Apps Motion

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Micro Hack 2 — SDC Workloads Workbook (Participants)

Track: **SDC Workloads (Fabric Extensibility Toolkit)** Scenario: **GreenGrid Analytics** (fictional SDC)

Print-ready PDF: [microhack-2-sdc-workloads-workbook.pdf](#)

This is a **didactic** micro-hack. You are **not** expected to *find* the solution — every step tells you **exactly what to type or paste, why, and what you should see**. Follow them in order and you get a working Fabric workload, starting from a clean machine.

Day agenda (1 day)

Morning — presentations & demos. Afternoon — hands-on hack. This is a **standalone one-day micro-hack** for the SDC Workloads (Extensibility Toolkit) track.

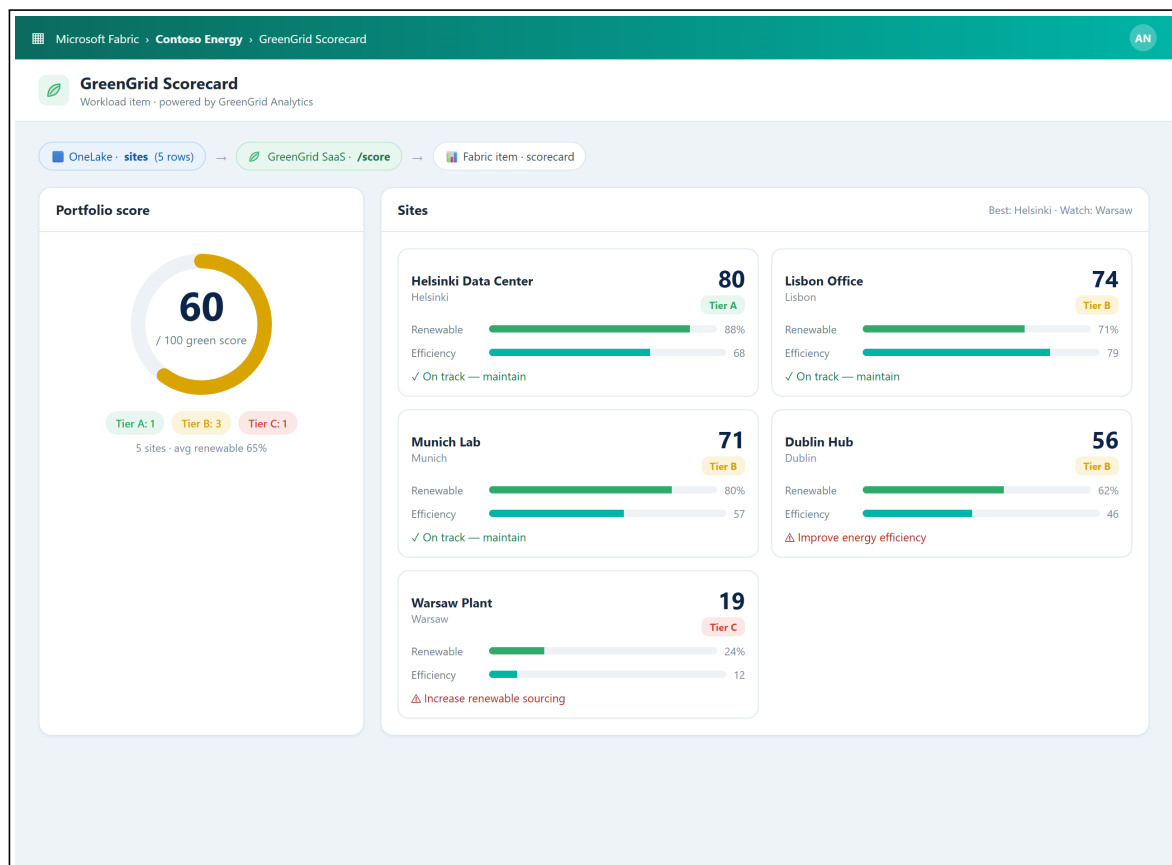
Time	Session
09:00	Welcome & motion overview — the Fabric apps motion for EMEA
09:30	Microsoft Fabric foundations — OneLake, capacity, workspaces
10:00	Extensibility Toolkit foundations — workloads, items, Dev Gateway
10:30	Break
10:45	Demo · GreenGrid Scorecard — the workload you'll build + the SaaS prerequisite
11:15	The hack brief, teams & environment check
12:00	Lunch
13:30	Hack · Sprint 1 — clone, run Hello World, render with fake data (steps 1–7)
15:30	Break
15:45	Hack · Sprint 2 — make it graphical (step 9); *optionally* switch to OneLake (step 8)
16:45	Team demos (5 min each)
17:00	Wrap-up, KPIs & next steps

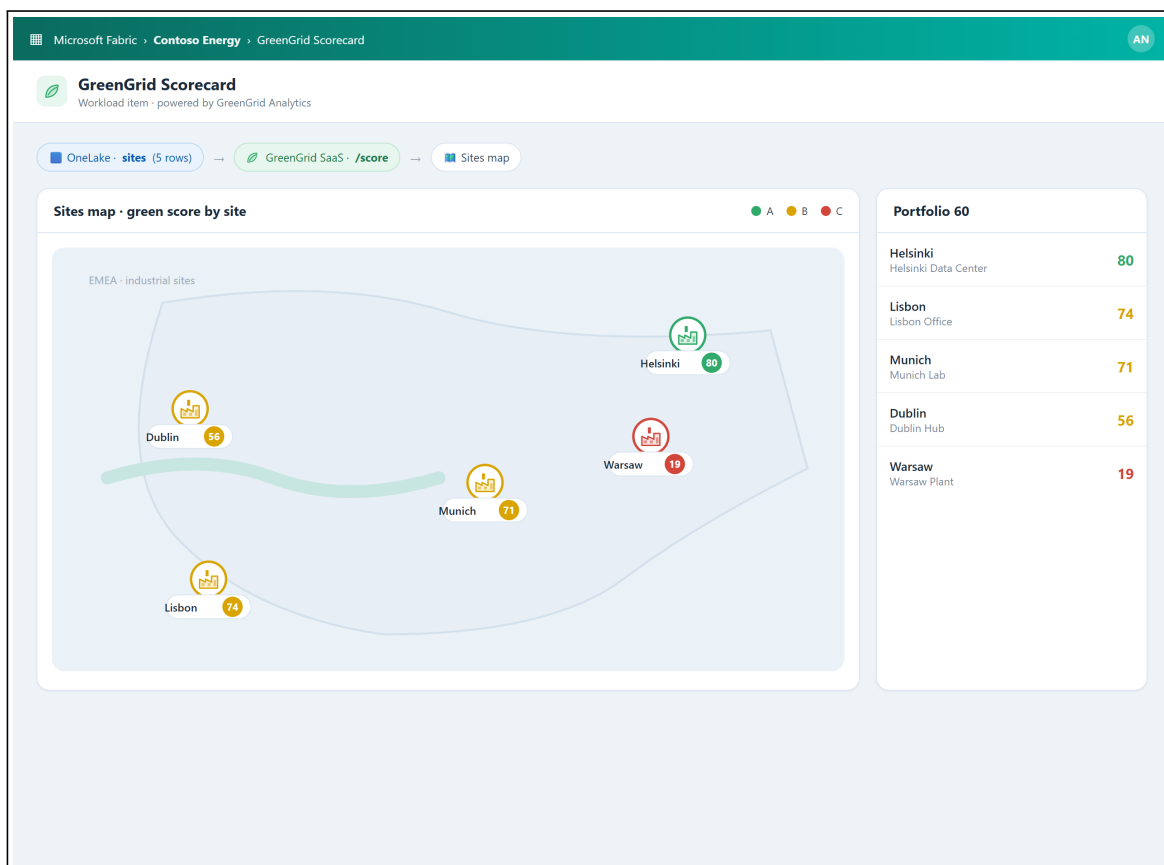
Part A — Understand

1) The scenario in plain words

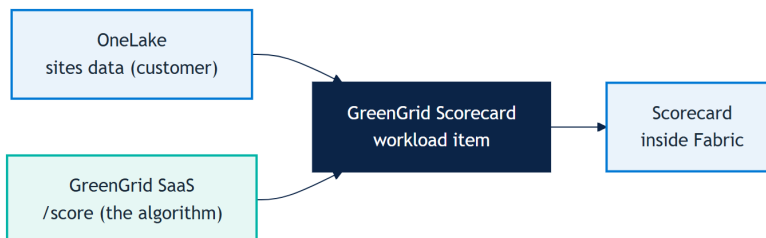
GreenGrid Analytics is a software company (an **SDC** — a solution development company). Its product scores how **sustainable** a customer's sites are, from their energy use and renewable mix. The customer's data already lives in Microsoft Fabric (**OneLake**). Your job: bring GreenGrid's scoring **inside Fabric**, as a native screen, **without moving the data**.

The result looks like this — a sustainability scorecard and a map of the industrial sites, rendered inside Fabric:





2) The three pieces (and who owns the algorithm)



- **OneLake** = where the **customer's data** lives.
- **GreenGrid SaaS** = a small web service that **holds the scoring algorithm** (GreenGrid's IP). *The trainer usually hosts it; if you run solo, you start it yourself in Step 6a — `cd src/workLoadsdsc && npm run saas:start`.*
- **The workload** = the piece **you build today**: a Fabric *item* that reads OneLake, calls the SaaS, and draws the scorecard.

3) Key words (30-second primer)

- **Workload**: a web app that runs *inside* the Fabric portal. You host it, Fabric shows it.
- **Item**: one thing a user creates in a workspace (here: a *GreenGrid Scorecard*).
- **Extensibility Toolkit**: Microsoft's open-source kit + SDK to build such workloads.
- **Dev Gateway**: a local bridge so Fabric can show your workload while it runs on **your machine** — this is what makes a one-day hack possible.
- **OBO token** (On-Behalf-Of): a secure token the workload gets so it can read the user's OneLake data **as that user** — no passwords, no data copies.

4) Prerequisites & environment setup (do this first)

Modeled on Microsoft's [Build LAB514](#) setup. On a lab VM most tools are pre-installed.

Tools you need:

- Node.js 20+ and npm
- **PowerShell 7**, **.NET (x64)**, **Azure CLI** (the toolkit setup script uses them)
- Visual Studio Code
- **GitHub Copilot CLI** (the copilot terminal agent) — used for the graphical polish step

Sign in (two accounts):

1. **GitHub Copilot** — run `copilot`, then `/login`, choose **GitHub.com**, finish the browser sign-in, confirm “Signed in successfully”, then `/exit`.
2. **Microsoft Fabric** — open `https://app.fabric.microsoft.com` and sign in. (If it shows “Power BI” bottom-left, switch it to “Fabric”.)

From the trainer / your tenant:

- A **Fabric workspace** with a **capacity** assigned (where the workload runs).
- Permission to create an **Entra app** (or one provided) — required even with the Dev Gateway; `Setup.ps1` creates it.
- The **GreenGrid SaaS URL + API key**.

Verify your setup:

```
node --version
npm --version
copilot --version
pwsh --version
```

What you should see. All print a version, you’re signed in to GitHub Copilot and Fabric, and you have the SaaS URL + key in hand.

5) Team framing (fill this in 2 minutes, then build)

- Team name:
- SaaS URL + API key:
- One sentence you’ll say in the demo:

Part B — Build it step by step (full solution)

Golden rule (read this!): first get the toolkit running with the built-in **Hello World** (steps 1–4). Only then build GreenGrid (steps 5–9). And inside GreenGrid, render **fake data first** (step 7), *then* switch to real OneLake data (step 8). Never debug security before something shows on screen.

Every step says **what it is**, **what to type/paste**, and **what you should see**.

Step 1 — Clone the toolkit

What this is. Get Microsoft’s Extensibility Toolkit on your machine. This is a real repository — you clone it and start from it.

```
git clone https://github.com/microsoft/fabric-extensibility-toolkit
cd fabric-extensibility-toolkit
```

What you should see. A `fabric-extensibility-toolkit` folder with `Workload/` and `scripts/` inside.

Step 2 — Configure the dev environment (one script)

What this is. A setup script that creates the Entra app, writes your `.env`, and downloads the Dev Gateway. It asks you to sign in and pick your workspace.

```
cd scripts/Setup
./Setup.ps1 -WorkloadName "Org.GreenGrid"
```

The WorkloadName must look like Organization.Name — use Org.GreenGrid for the hack. On macOS/Linux run `pwsh ./Setup.ps1 -WorkloadName "Org.GreenGrid"`.

Why an Entra app even with the Dev Gateway? The Dev Gateway only **routes** your local workload into Fabric — it does **not** provide identity. Your workload authenticates with **Microsoft Entra**, and the Entra app is what lets it receive Fabric tokens and acquire the **OBO token** to read OneLake (step 8). Setup.ps1 creates this app for you (or reuses an existing one).

What you should see. The script ends with “setup complete” and prints the next steps.

Step 3 — Start the dev server + the Dev Gateway

What this is. Two long-running processes, in **two terminals**. One serves your workload UI from localhost; the other connects Fabric to that localhost.

```
# Terminal 1 – the frontend + APIs
cd scripts/Run
./StartDevServer.ps1
```

```
# Terminal 2 – the bridge to Fabric
cd scripts/Run
./StartDevGateway.ps1
```

Then, in the **Fabric portal**: open **Settings -> Admin portal** and enable the developer tenant settings, then **Settings -> Developer settings -> enable Fabric Developer Mode**.

What you should see. Both terminals keep running without errors, and Developer Mode is ON in Fabric.

If StartDevGateway.ps1 fails with Dev instance registration ... Forbidden, error-Code: FeatureNotAvailable : your tenant hasn't enabled customer-developed workloads. An **admin** must turn on, under **Admin portal -> Tenant settings -> Additional workloads: Capacity admins and contributors can add and remove additional workloads, Workspace admins can develop workloads, and Users can see and work with additional workloads not validated by Microsoft** — **and** the workspace must be on a **Fabric/Trial capacity** (not Pro/PPU). Full steps in the setup guide (§6.1). It is not a code issue — the gateway is refused by Fabric.

Not a tenant admin? You can't self-enable it — either ask your admin for those toggles, run the hack in a tenant **you** administer (a Fabric Trial / demo tenant), or keep building locally without Fabric: `npm run saas:start + npm run demo:build`, then open `src/workloadsdc/demo/out/scorecard.html` (same algorithm, no Fabric needed). See setup §6.1.1.

Step 4 — Hello World test (prove the gateway works, before building anything)

What this is. The toolkit ships a **Hello World** item. Create it in Fabric to confirm the whole chain — *your localhost code -> Dev Gateway -> Fabric* — works. **This is your safety checkpoint.**

1. Go to <https://app.fabric.microsoft.com/workloadhub/detail/Org.GreenGrid.Product?experience=developer> (the workload shows as “Hello Fabric!” until you rename it).
2. Click the **Hello World** item type on the left.
3. Pick your **dev workspace** in the dialog.
4. The Hello World editor opens.

What you should see. The Hello World editor renders **inside Fabric**, served from your machine. If you see it, everything works — now you can build GreenGrid. (*Nothing shows up? Developer Mode is off, or one of the two terminals isn't running.*)

Step 5 — Create the GreenGrid item

What this is. Generate a fresh item to hold your scorecard (don't overwrite Hello World).

```
cd scripts/Setup
./CreateNewItem.ps1 -Name "GreenGridScorecard"
```

This creates Workload/app/items/GreenGridScorecardItem/ on your machine, including **GreenGridScorecardItemEditor.tsx** — the **React component Fabric renders when your item is opened** (this is “the editor”; it starts nearly empty).

CreateNewItem.ps1 scaffolds the code but leaves 4 things for you to wire up — do all four now, or the Dev Gateway rejects the item / Fabric won't surface it.

1. Register it in ITEM_NAMES (else it isn't packaged -> not in the manifest). Edit **all three** Workload/.env.dev, .env.test, .env.prod and append the -Name value:

```
- ITEM_NAMES=HelloWorld
+ ITEM_NAMES=HelloWorld,GreenGridScorecard
```

2. Add the locale keys (else Missing keys in en-US locale: ..._DisplayName, ..._DisplayName_Plural). In Workload/Manifest/assets/locales/en-US/translations.json add three keys (Display/Plural are used by the item manifest, Description by the create-card in step 4):

```
"GreenGridScorecardItem_DisplayName": "GreenGrid Scorecard",
"GreenGridScorecardItem_DisplayName_Plural": "GreenGrid Scorecards",
"GreenGridScorecardItem_Description": "Score the sustainability of your
  ↵ sites in Fabric."
```

Every key must be **referenced somewhere** — an unreferenced key gives Unused keys in en-US locale.

3. Add the item icon (else MissingAssets: 'assets/images/GreenGridScorecardItem_Icon.png'). Drop a 48×48 PNG named <Item>_Icon.png into Workload/Manifest/assets/images/ (quickest fix: copy HelloWorldItem_Icon.png, swap a nicer icon later).

4. Add a create-card to Product.json (else the gateway registers fine and Hello World shows, but **your item never appears in New item**). In Workload/Manifest/Product.json add your item to homePage.recommendedItemTypes **and** add a card to createExperience.cards mirroring the HelloWorld one — set title/description/icon to your GreenGridScorecardItem_* keys/asset, availableIn: ["home", "create-hub", "workspace-plus-new", "workspace-plus-new-teams"], and "itemType": "GreenGridScorecard".

Restart both terminals so the manifest is rebuilt and re-delivered to Fabric — the Dev Server (Ctrl+C, ./StartDevServer.ps1) **and** the Dev Gateway (Ctrl+C, ./StartDevGateway.ps1). On restart the gateway should log Configured items from ITEM_NAMES: HelloWorldItem, GreenGridScorecardItem and Registering dev instance... should **succeed** (no fail).

Now **create an instance** of your item *in the Fabric portal* so you have something to render into:

1. **Refresh** Fabric (F5), then open the Workload Hub: <https://app.fabric.microsoft.com/workloadhub/detail/developer> (same place as Hello World in Step 4).
2. Click the **GreenGridScorecard** item type -> pick your **dev workspace** -> the editor opens.

Type vs instance. CreateNewItem.ps1 scaffolds the item **type + its editor code** (on your machine). Creating it in Fabric makes an **instance** that runs that code inside Fabric. Keep this instance open: every time you save the editor .tsx, the dev server hot-reloads and this open item re-renders — you do **not** re-create the item for each change.

Don't see GreenGridScorecard in “New item”? 99% of the time it's the ITEM_NAMES step above: confirm the item is added in **all three** .env files, restart **both** the Dev Server **and** the Dev Gateway, then **refresh** Fabric. (*The BuildManifestPackage ... CursorPosition: The handle is invalid warning is harmless — it's a console-cursor quirk; Delivered manifest package successfully means the build is fine.*)

What you should see. An empty GreenGridScorecard editor opens in Fabric (your code, rendered in the portal).

Step 6 — Start & check the GreenGrid SaaS

What this is. The SaaS holds GreenGrid's scoring algorithm — your workload is useless without it, so it must be **running and reachable** before you build.

6a. Start the SaaS

If the **trainer hosts it**, just grab the **SaaS URL + API key** they give you and skip to 6b.

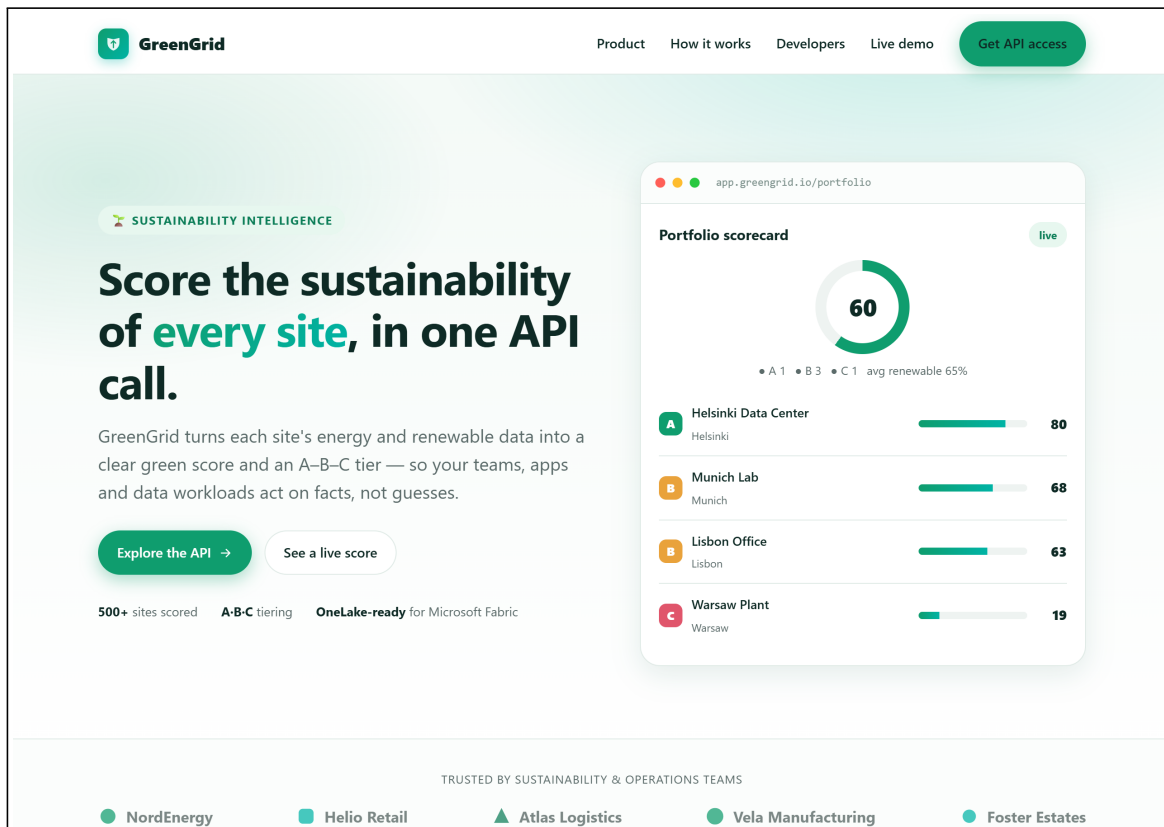
If you're **running it yourself** (solo, or as a local fallback), start it from the kit:

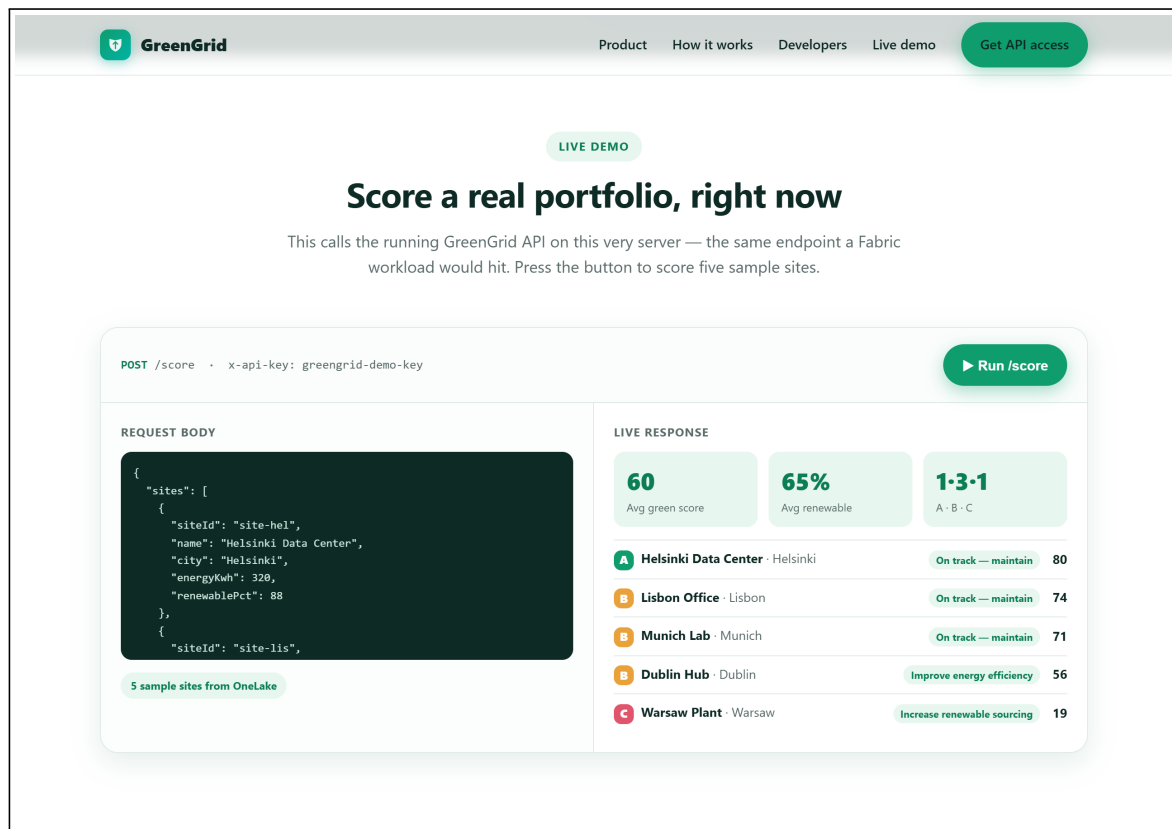
```
cd src/workloadsdc
npm install
npm run saas:start
```

It starts on **http://localhost:8787** with the default API key **greengrid-demo-key** (override with the `GREENGRID_API_KEY` env var). The console should print:

```
GreenGrid SaaS listening on http://localhost:8787
GET / (marketing site + live API demo)
GET /health
POST /score (header: x-api-key: greengrid-demo-key)
```

Keep this terminal open — the SaaS must stay running for the whole hack. Open **http://localhost:8787/** in a browser to see the GreenGrid website + a **live /score demo** (same algorithm your workload will call).





6b. Check it's alive

Replace `<SAAS_URL>` with your URL (local: `http://localhost:8787`) and `<API_KEY>` with your key (local: `greengrid-demo-key`):

```
curl -i <SAAS_URL>/health # expect: 200
curl -s -X POST <SAAS_URL>/score \
  -H "Content-Type: application/json" -H "x-api-key: <API_KEY>" \
  -d '{"sites":[{"siteId":"s1","name":"Helsinki
    ↪ DC","city":"Helsinki","energyKwh":320,"renewablePct":88}]}'
```

On Windows PowerShell, `curl` is an alias for `Invoke-WebRequest` and won't accept the flags above — use `curl.exe` (real curl), or the native form:

```
Invoke-RestMethod http://localhost:8787/health
Invoke-RestMethod -Method Post -Uri http://localhost:8787/score `
  -Headers @{ 'x-api-key' = 'greengrid-demo-key' } -ContentType
    ↪ 'application/json' `
  -Body '{"sites":[{"siteId":"s1","name":"Helsinki
    ↪ DC","city":"Helsinki","energyKwh":320,"renewablePct":88}]}'
```

What you should see. `/health` returns 200, and `/score` returns a greenScore and a tier (Helsinki -> greenScore 80, tier A). Note your working **SAAS_URL + API_KEY** — you'll paste them into the workload client in Step 7.

Step 7 — Make it work with fake data (Milestone M1)

Create **four files** in `Workload/app/items/GreenGridScorecardItem/`, then render the scorecard from the generated editor. This proves the whole chain *screen -> SaaS -> screen*.

7a. contracts.ts — the data shapes

```
export type SiteRecord = {
  siteId: string; name: string; city: string; energyKwh: number; renewablePct:
  ↪ number;
};
export type GreenTier = 'A' | 'B' | 'C';
export type ScoredSite = SiteRecord & {
  efficiency: number; greenScore: number; tier: GreenTier; tip: string;
};
export type ScoreResponse = {
  sites: ScoredSite[];
  summary: { totalSites: number; avgScore: number; avgRenewablePct: number;
    tierCounts: Record<GreenTier, number>; best: ScoredSite; worst: ScoredSite; };
};
```

7b. greengridClient.ts — call the SaaS

```
import type { ScoreResponse, SiteRecord } from './contracts';

const SAAS_URL = 'http://localhost:8787'; // <- replace with the trainer URL
const API_KEY = 'greengrid-demo-key'; // <- replace with the trainer key

export async function scorePortfolio(sites: SiteRecord[]): Promise<ScoreResponse> {
  const res = await fetch(`${SAAS_URL}/score`, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json', 'x-api-key': API_KEY },
    body: JSON.stringify({ sites }),
  });
  if (!res.ok) throw new Error(`GreenGrid SaaS error ${res.status}`);
  return res.json();
}
```

7c. seed.ts — fake sites to start

```
import type { SiteRecord } from './contracts';
export const seedSites: SiteRecord[] = [
  { siteId: 'site-hel', name: 'Helsinki Data Center', city: 'Helsinki', energyKwh:
  ↪ 320, renewablePct: 88 },
  { siteId: 'site-lis', name: 'Lisbon Office', city: 'Lisbon', energyKwh: 210,
  ↪ renewablePct: 71 },
  { siteId: 'site-muc', name: 'Munich Lab', city: 'Munich', energyKwh: 430,
  ↪ renewablePct: 80 },
  { siteId: 'site-dub', name: 'Dublin Hub', city: 'Dublin', energyKwh: 540,
  ↪ renewablePct: 62 },
  { siteId: 'site-waw', name: 'Warsaw Plant', city: 'Warsaw', energyKwh: 880,
  ↪ renewablePct: 24 },
];
```

7d. Scorecard.tsx — the screen

It takes a `getSites` function, so swapping the data source later is **one line**.

```
import React, { useEffect, useState } from 'react';
import { FluentProvider, webLightTheme, Spinner, Badge } from
  ↪ '@fluentui/react-components';
import { scorePortfolio } from './greengridClient';
import type { ScoreResponse, SiteRecord } from './contracts';
```

```

const tierColor = (t: 'A' | 'B' | 'C') => (t === 'A' ? '#2faa6a' : t === 'B' ?
  ↪ '#d9a300' : '#d1453b');

export function Scorecard({ getSites }: { getSites: () => Promise<SiteRecord[]> }) {
  const [data, setData] = useState<ScoreResponse | null>(null);
  const [error, setError] = useState<string | null>(null);

  useEffect(() => {
    getSites().then(scorePortfolio).then(setData).catch((e) =>
  ↪ setError(String(e)));
  }, [getSites]);

  if (error) return <div role="alert">Failed: {error}</div>;
  if (!data) return <Spinner label="Scoring sites..." />;

  return (
    <FluentProvider theme={webLightTheme}>
      <div style={{ padding: 24 }}>
        <h1>GreenGrid Scorecard</h1>
        <p>Data from OneLake, scored by GreenGrid</p>
        <div style={{ fontSize: 48, fontWeight: 700 }}>{data.summary.avgScore}<span
  ↪ style={{ fontSize: 18 }}> / 100</span></div>
        <div style={{ display: 'grid', gridTemplateColumns: 'repeat(2,1fr)', gap:
  ↪ 12, marginTop: 16 }}>
          {data.sites.map((s) => (
            <div key={s.siteId} style={{ border: '1px solid #dbe6f1', borderRadius:
  ↪ 12, padding: 14 }}>
              <div style={{ display: 'flex', justifyContent: 'space-between' }}>
                <strong>{s.name}</strong>
                <Badge appearance="filled" style={{ background: tierColor(s.tier)
  ↪ }}>Tier {s.tier}</Badge>
              </div>
                <div style={{ color: '#7a8a9a', fontSize: 12 }}>{s.city}</div>
                <div style={{ fontSize: 26, fontWeight: 700 }}>{s.greenScore}</div>
                <div style={{ fontSize: 13 }}>{s.tip}</div>
              </div>
            </div>
          ))}
        </div>
      </FluentProvider>
    );
  }

```

7e. Render it from the generated editor

What this is. Steps 7a–7d created the building blocks (types, SaaS client, seed data, the Scorecard component) — but nothing displays them yet. 7e **wires Scorecard into the item's editor** so it shows inside Fabric. This is the step that turns files-on-disk into a visible item.

The scaffold splits the editor into **views**: `GreenGridScorecardItemEmptyView.tsx` and `GreenGridScorecardItemDefaultView.tsx` (the main view). You (1) register the editor's **route**, (2) render your Scorecard in the `DefaultView`'s center, and (3) make the item open that view:

1. In **App.tsx** — register the editor route, or the item opens to a **blank page** (Fabric navigates to `/<Item>-editor/:itemObjectId` but no `<Route>` matches -> nothing renders, *with no error in the console*). `CreateNewItem.ps1` does **not** add this. Mirror the `HelloWorld` route:

```

import { GreenGridScorecardItemEditor } from "../items/GreenGridScorecardItem";
// ...inside <Switch>, next to the HelloWorld route:
<Route path="/GreenGridScorecardItem-editor/:itemObjectId">
  <GreenGridScorecardItemEditor workloadClient={workloadClient}
  ↪ data-testid="GreenGridScorecardItem-editor" />

```

```
</Route>
```

2. In **GreenGridScorecardItemDefaultView.tsx** — import your component + seed data and put the Scorecard in the center slot (you can keep or remove the left “Getting Started” panel):

```
import { Scorecard } from "../Scorecard";
import { seedSites } from "../seed";
// ...
center={{ content: <Scorecard getSites={() => Promise.resolve(seedSites)} /> }}
```

3. In **GreenGridScorecardItemEditor.tsx** — by default the editor renders **no view** (also a blank page) because ItemEditor starts with `currentView = initialView || null`. Add **initialView** so it opens the scorecard view directly:

```
<ItemEditor
  isLoading={isLoading}
  loadingMessage={...}
  initialView={EDITOR_VIEW_TYPES.DEFAULT} // ← without it the page is blank
  ribbon={...}
  views={views}
/>
```

Blank page with no console error? It's almost always **#1 (the missing App.tsx route)** — the console shows a navigation to `/<Item>-editor/...` but nothing renders. Then **#3** (no `initialView`).

Save. The dev server hot-reloads; refresh your open GreenGrid item in Fabric (keep the **SaaS** at `http://localhost:8787` and both Dev Server + Dev Gateway running). (If you closed the item, reopen it from the Workload Hub as in Step 5.)

What you should see — Milestone M1. The item shows the portfolio score and 5 site cards with their tier, inside Fabric.

Fake data or real backend? Both — that's the point of M1. The **5 sites are fake** (`seedSites`; they become real OneLake data in Step 8), but the **scoring is a real call to the SaaS** you started: Scorecard POSTs the sites to `http://localhost:8787/score`, which runs the GreenGrid algorithm and returns the scores you see. Verify it: **F12 -> Network** shows a score request (200) to `localhost:8787`, and the **SaaS terminal logs** `POST /score - 5 site(s) -> avg ....` Stop the SaaS and refresh -> the card shows Failed: ..., proving it depends on the backend.

Step 8 — Switch to real OneLake data (Milestone M2) — optional, do it at the end

Optional — do this last (or skip for now). M1 (Step 7e) already proves the full screen -> SaaS -> screen chain, and **Step 9 works on seed data too**, so you can **jump straight to Step 9** and come back to wire OneLake at the end (or if time allows). The only thing Step 8 changes is *where the input sites come from* (seed array -> OneLake CSV) — one line in the view.

What this is. Replace the fake `seedSites` with the customer's **real** site data in OneLake.

Where does the data live?

In Fabric, customer data lives in a **Lakehouse** inside OneLake. A Lakehouse has **Files** (raw files: CSV, Parquet...) and **Tables** (Delta). For this hack the sites are a **CSV file**: `Files/sites.csv`. (A CSV is the simplest thing to read from a workload; a Delta table would need a SQL endpoint — out of scope for one day.)

Format of Files/sites.csv (a ready copy is in the kit at `src/workloadsgdc/data/sites.csv`):

```
siteId,name,city,energyKwh,renewablePct
site-hel,Helsinki Data Center,Helsinki,320,88
site-lis,Lisbon Office,Lisbon,210,71
site-muc,Munich Lab,Munich,430,80
site-dub,Dublin Hub,Dublin,540,62
site-waw,Warsaw Plant,Warsaw,880,24
```

The trainer pre-loads this file (Lakehouse -> **Files** -> **Upload**). To test your own, upload a CSV with the **same header** to Files/.

You also need two IDs (visible in the Lakehouse URL, or from the item context): `workspaceId` and `lakehouseId`.

8a. onelake.ts — read & parse the CSV

```
import type { WorkloadClientAPI } from '@ms-fabric/workload-client';
import type { SiteRecord } from './contracts';

const ONELAKE_DFS_BASE = 'https://onelake.dfs.fabric.microsoft.com';
const ONELAKE_STORAGE_SCOPE = 'https://storage.azure.com/user_impersonation';

/**
 * Reads sites.csv from a Lakehouse "Files" area in OneLake, as the signed-in user.
 * Expected CSV header: siteId,name,city,energyKwh,renewablePct
 */
export async function readSites(
  client: WorkloadClientAPI,
  workspaceId: string,
  lakehouseId: string,
): Promise<SiteRecord[]> {
  // 1) Token to read OneLake AS the signed-in user (OBO).
  const { token } = await client.auth.acquireFrontendAccessToken({
    scopes: [ONELAKE_STORAGE_SCOPE],
  });

  // 2) Read the CSV file from the Lakehouse "Files" area.
  const url = `${ONELAKE_DFS_BASE}/${workspaceId}/${lakehouseId}/Files/sites.csv`;
  const res = await fetch(url, { headers: { Authorization: `Bearer ${token}` } });
  if (!res.ok) throw new Error(`OneLake read failed: ${res.status}`);

  // 3) Parse CSV text into rows.
  const [header, ...lines] = (await res.text()).trim().split(/\r?\n/);
  const cols = header.split(',').map((c) => c.trim());
  return lines.filter(Boolean).map((line) => {
    const cells = line.split(',');
    const row = Object.fromEntries(cols.map((c, i) => [c, (cells[i] ?? '').trim()]));
    return { siteId: row.siteId, name: row.name, city: row.city,
      energyKwh: Number(row.energyKwh), renewablePct: Number(row.renewablePct) };
  });
}
```

Use the right auth API. The OBO token comes from `client.auth.acquireFrontendAccessToken({ scopes: [...] })` — *not* `acquireAccessToken({ additionalScopesToConsent })`, which **does not exist** on the toolkit client and will fail to compile. A compile error here breaks the whole bundle, so the **editor opens blank with no console error** (same symptom as a missing route). If `npx tsc --noEmit` isn't green, fix it before reloading Fabric.

Paste ASCII only — beware copy-paste artifacts. If the item opens blank right after adding `onelake.ts`, check for **non-ASCII characters** that slip in from chat/PDF copy: a Unicode hyphen - (U+2010) in the package name `@ms-fabric/workload-client` -> *"Cannot find module"*; or stray glyphs like `'(U+230B)` / `□'(U+21AA)` and hard line breaks inside the URL template literal -> *"Unterminated template literal"* and a cascade of TS errors. Re-type the dashes/back-ticks by hand, or run a quick find for any character outside the ASCII range.

8b. Swap one line in the editor

In `GreenGridScorecardItemDefaultView.tsx` (where you put the Scorecard in 7e), change the center line to read from `OneLake`. `workloadClient` is already a prop of this view; for the IDs, add `item` back to the props and use `item.workspaceId` / the lakehouse id (from the item or the selected lakehouse):

```
import { readSites } from './onelake';

// before (fake):
center={{ content: <Scorecard getSites={() => Promise.resolve(seedSites)} /> }}

// after (real OneLake CSV):
center={{ content: <Scorecard getSites={() => readSites(workloadClient,
  ↪ workspaceId, lakehouseId)} /> }}
```

What you should see — Milestone M2. The same scorecard, now built from the **OneLake CSV**. (If the OBO token fights you, keep the fake-data line as a fallback and keep moving.)

Blank vs “Failed: ...” — two different problems. A **blank** editor means the bundle didn't compile (a TypeScript error in `onelake.ts` — wrong auth API or a non-ASCII character; see 8a). If instead the Scorecard renders but shows “**Failed: ...**”, the build is fine and you're hitting a **runtime** OneLake issue (token consent, the CSV missing at `Files/sites.csv`, a 403, or CORS) — debug the fetch, not the build.

The Sites map still uses seed data. This swap only changes the **Scorecard** view's input. The **Sites map** tab keeps rendering `seedSites` until you wire it to `readSites` the same way — optional polish, do it only if you have time.

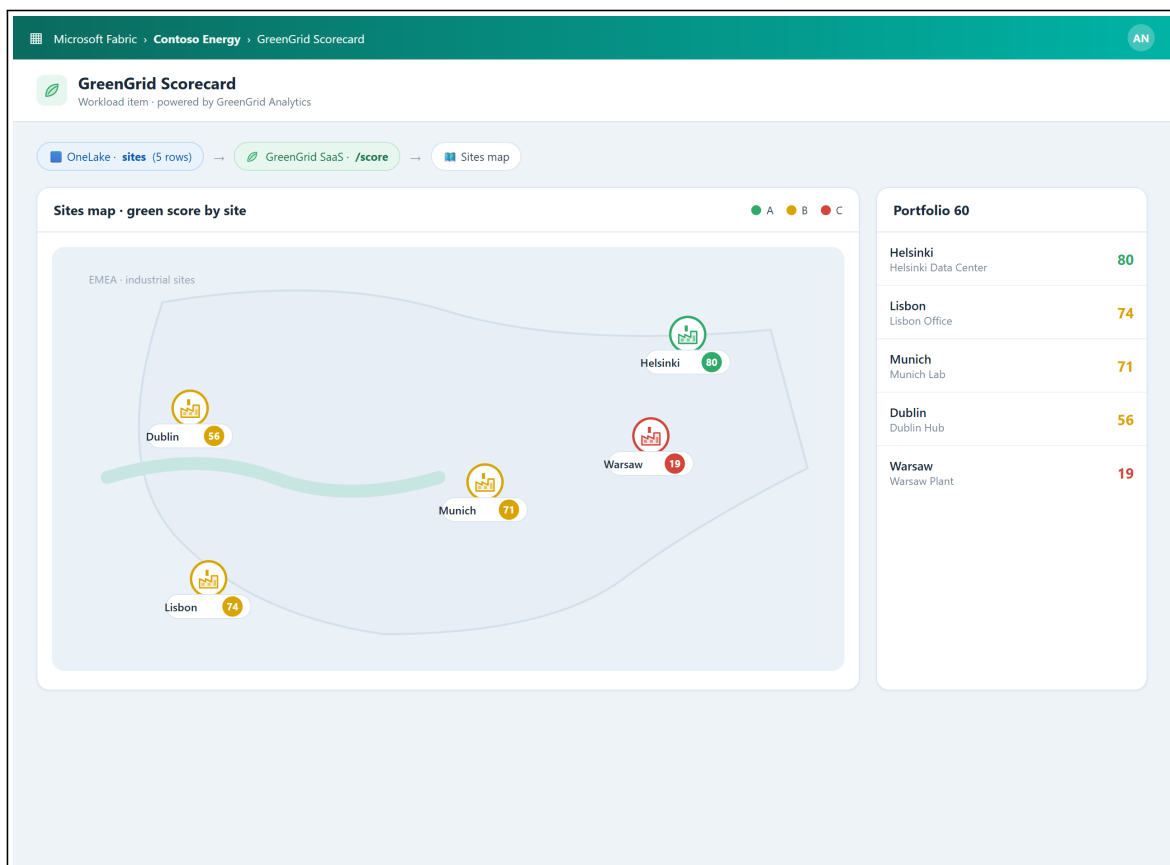
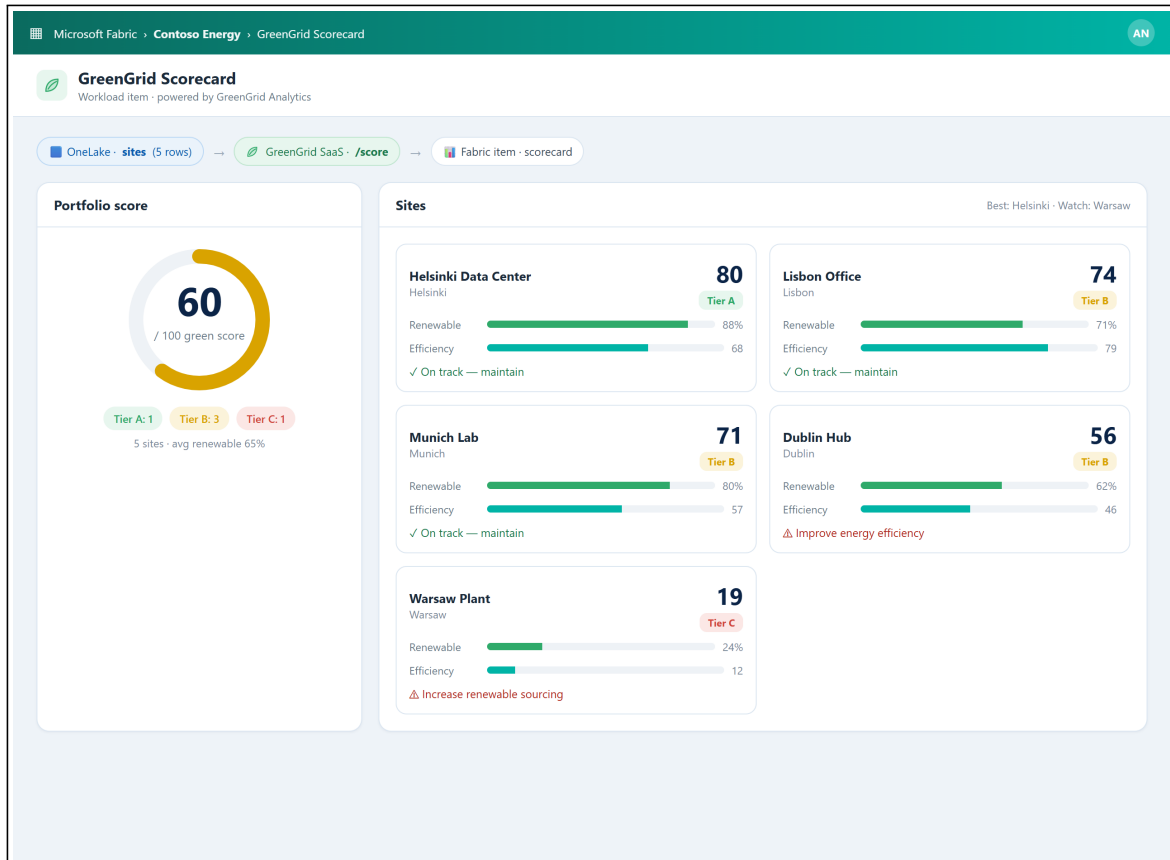
Step 9 — Make it graphical (Milestone M3)

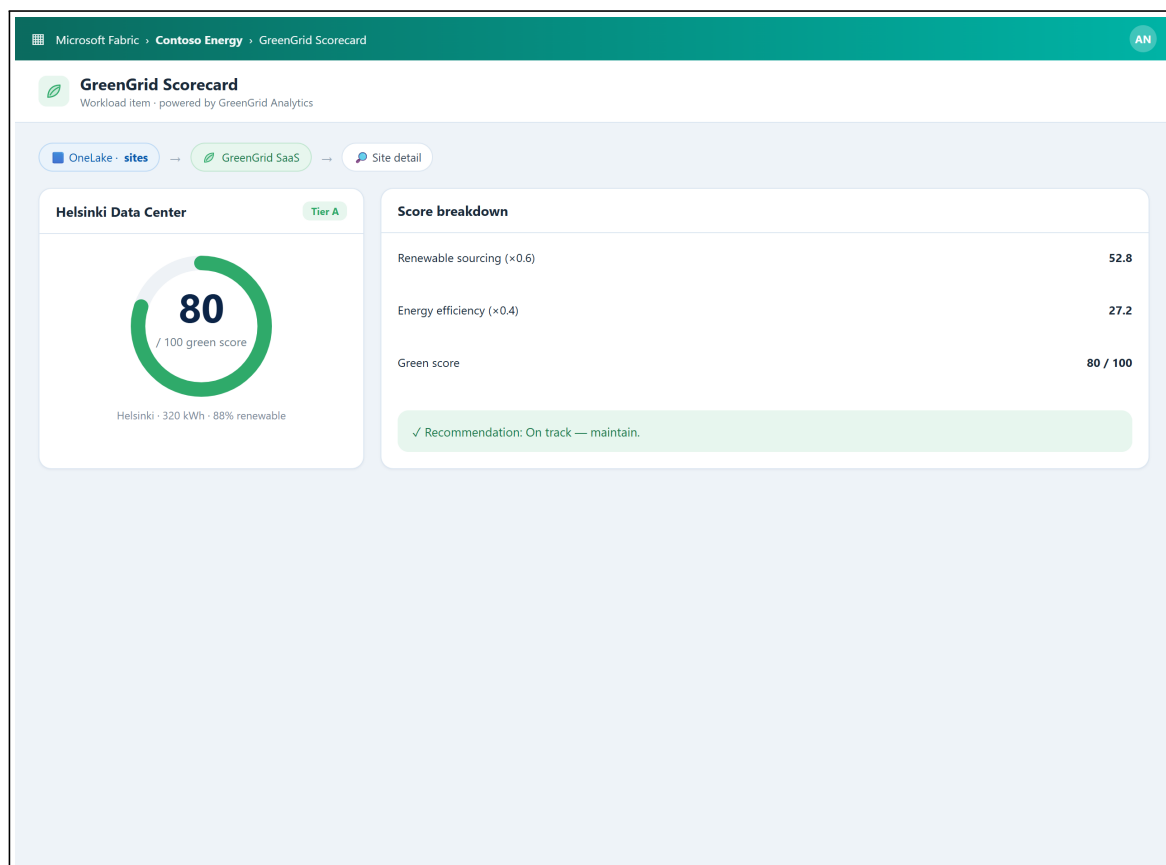
What this is. Upgrade the plain list into the polished scorecard + map. Easiest with GitHub Copilot (the toolkit ships AI instructions in `.github/copilot-instructions.md`):

Upgrade the GreenGrid Scorecard screen: add a circular green-score gauge for the
 ↪ portfolio,
 an A/B/C tier distribution, provenance chips "Data from OneLake" then "Scored by
 ↪ GreenGrid",
 and a second view showing the sites as factory markers on a simple map, colored by
 ↪ tier.
 Style: Fluent UI, teal #00b4a6 + green accents. Keep the `getSites` data flow.

Where do I run Copilot, and do I restart anything? - **Directory:** run Copilot in the **fabric-extensibility-toolkit repo** (your workload front-end) — open that folder in VS Code, or run the GitHub Copilot CLI **from that folder**. It edits the React/TSX under `Workload/app/items/GreenGridScorecardItem/` (`Scorecard.tsx`, `GreenGridScorecardItemDefaultView.tsx`). **Not** the SaaS repo, **not** the FY27 motion repo. - **After the prompt:** small edits to *existing* .tsx are picked up by **Dev Server hot reload** — just refresh the Fabric item (Ctrl+Shift+R). But if Copilot **added new files or a new view** (likely for the gauge/map), **restart the Dev Server** so the new modules get bundled: Ctrl+C in its terminal, then `cd scripts\Run && .\StartDevServer.ps1`; then refresh Fabric. Keep the **SaaS** and the **Dev Gateway** running. - You only restart the **Dev Gateway** for *manifest/back-end* changes (`ITEM_NAMES`, `Product.json`, locale keys, item icon) — **never** for UI code changes. - **Keep the `getSites` data flow** (as the prompt says) so the SaaS call still happens — only the *visuals* change, not the chain. - **Restarted the Dev Server but Fabric still shows the old UI?** Two layers of cache: 1. **Build cache** — webpack keeps a persistent cache at `Workload/node_modules/.cache` that can serve **stale compiled modules** even after a restart (especially after a big change like the Copilot upgrade). **Delete `Workload/node_modules/.cache`** then restart the Dev Server (the rebuild is slower — wait for compiled successfully). 2. **Browser/iframe cache** — then **F12 -> Network -> tick “Disable cache”** and reload, or right-click reload -> **“Empty Cache and Hard Reload”**, or **close & reopen the item** (or Incognito). A plain Ctrl+Shift+R often isn't enough.

What you should see — Milestone M3. A gauge, A/B/C tiers and the sites map, rendered natively inside Fabric:





Wrap-up

Demo storyboard (5 minutes)

1. **Context** (30s): the SaaS value (the algorithm) you brought into Fabric.
2. **Flow** (3m): open the item, read OneLake, call the SaaS, show the scorecard + map.
3. **Architecture** (45s): OneLake (data) + SaaS (algorithm) + workload (native UI).
4. **Publish plan** (45s): from Dev Gateway to a released workload.

Reflection

- Where did the OneLake -> SaaS contract surprise you?
- What stays in the SaaS (the IP) vs. in the workload (the experience)?
- What would you harden before production?

Reference assets

- Full solution code: `src/workloadsgdc/`
- SaaS prerequisite: `src/workloadsgdc/saas/`
- Sample OneLake data: `src/workloadsgdc/data/sites.csv`
- Trainer answer key + screenshots: `docs/microhack-2-sdc-workloads-solution.md`